

Yash Raj Agarwal

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Guwahati, Assam, India

EDUCATION

Manipal Institute of Technology

Bachelor of Technology in Computer Science & Engineering; CGPA: 7.65/10.0

July 2024 – July 2028

The Assam Valley School

High School (12th Grade), ISC; Percentage: 87.6%

Manipal, Karnataka, India

April 2021 – April 2023

Tezpur, Assam, India

EXPERIENCE

Froker

Frontend Developer, Freelance

May 2025 - July 2025

- Collaborated remotely with a large tech team to help scale a **MERN-stack** app from a few hundred to **50,000+ downloads**, gathering and translating requirements into interface specifications and high-fidelity prototypes
- Built **React** frontend features across **20+ client projects** from requirement discussions through deployment, driving repeat engagements through consistent client trust

Universal Systems

Software Developer Intern

May 2026 - July 2026

- Designed and deployed a client-facing business website end-to-end using **Astro**, **TypeScript**, **HTML**, **CSS**, achieving **100/100 Lighthouse Performance and SEO score**
- Built a business analytics dashboard using **Python**, **Pandas**, **NumPy** to process daily inventory and business data for EDA, trend analysis, inventory insights and statistical reporting to support data-driven decisions

PROJECTS

VaaniPay | *Python, FastAPI, Twilio, Sarvam AI, bcrypt*

[Github Link](#)

- Designed a voice-first **UPI payments** solution for financial inclusion, built on a **Twilio**-based **IVR** system that lets feature-phone users authorize transactions over standard voice calls delivering **7+ offline banking workflows** with zero internet dependency for underserved users
- Engineered a fault-tolerant session recovery mechanism and secured transactions with **bcrypt MPIN** validation, integrating **Sarvam AI** to dynamically route **260+ voice interactions** across **9 regional languages**.

AeroHealth | *React, Node.js, Azure AI Language, Socket.IO, OpenRouteService*

[Github Link](#)

- Built an **AI-driven** healthcare routing platform dispatching patients to optimal hospitals in real time, using **Azure AI Language** to extract clinical entities from free-textsymptoms and classify across **11 departments and 5 urgency levels**, with a rule-based fallback ensuring the system degrades gracefully if Azure is unreachable
- Designed a **6-variable weighted forecasting model** predicting each hospital's **60-minute occupancy**, feeding a hospital-scoring engine that dynamically reroutes ambulances via **OSRM traffic-aware routing** when a destination saturates en route

NetSight AI | *FastAPI, Redis, WebSockets, Docker, Scapy*

[Github Link](#)

- Built a distributed, event-driven network monitoring platform (**Redis Pub/Sub**, **WebSockets**, **Docker**) processing **10,000+** real-time telemetry across **500+ nodes**, cutting manual triage time for anomaly detection
- Implemented **Scapy**-based packet inspection and threat-detection pipelines with an integrated **LLM diagnostics copilot** to automate root-cause analysis and shorten incident-response time

TECHNICAL SKILLS

Languages: Java, C++, C, Python, JavaScript, SQL

Frontend Development: React.js, Vite, HTML, CSS

Backend Development: REST APIs, Node.js, Express.js, Flask, Supabase

Databases: MySQL, PostgreSQL, MongoDB, Redis

Cloud, DevOps & Tools: Microsoft Azure, Vercel, Docker, n8n, Git, GitHub, Postman

Core CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

ACHIEVEMENTS

Top 50 - NameSpace Hackazards 2026 : Largest international hackathon with **31000+ participants**

Top 250 - Microsoft AI Unlocked Hackathon 2026

1st Place - ACM Hackathon 2026, Manipal: architected and pitched project VaaniPay

1st Place - iDesign, TechTatva 2025: Designed an AI-powered in-car infotainment system in Figma

1st Place - Negative Space 2026, IECSE UI/UX Competition: conceived an end-to-end college discovery platform